Joint NCTS/FGCPA-LeCosPA Meeting on Dark Energy

Study of Critical Gravity

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2 Critical Gravity

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1 Introduction

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2+1 Gravity

- The action of three-dimensional pure quantum gravity is

$$I = \frac{1}{16\pi G}\int \sqrt{-g}\, d^3x \left(R + \frac{2}{l^2}\right),$$

where the cosmological constant $\Lambda = -l^{-2}$ is negative with l is the radius of AdS_3 spacetime.

- Classically, $2+1$ dim pure gravity can be expressed in terms of gauge theory, the Chern-Simons gauge theory,

$$I = \frac{k}{4\pi}\int tr^* \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right)$$

with

$$k = \frac{-l}{4G}, \quad A = \begin{pmatrix} \omega & e/l \\ -e/l & 0 \end{pmatrix}$$

in $SO(2,2)$ group. [E. Witten (1998)]

- $I = k_L I_{CS_L} + k_R I_{CS_R}$ with one-form $A^{L/R} = \omega \mp {}^*e/l$ in $SO(2,1)_L \otimes SO(2,1)_R$ locally equivalent to $SO(2,2)$ group.

Topological Massive Gravity (TMG)

- In 2+1 dim gravity with Λ

$$\begin{cases} \Lambda > 0 & \text{do not know how defined mathematically observables,} \\ \Lambda = 0 & \text{only locally asymptotic to } M_4 \text{ at infinity} \Rightarrow \text{no S-matrix,} \\ \Lambda < 0 & \text{find a dual CFT to deal with } (AdS/\text{CFT.}) \end{cases}$$

- Adding high derivative Chern-Simons term to the wrong-sign Einstein-Hilbert action

$$I = \int \sqrt{-g}\, d^3x \left[\frac{-1}{2\kappa^2} R + \frac{1}{4\kappa^2 \mu} \epsilon^{\lambda\mu\nu} \Gamma^{\rho}_{\lambda\sigma} (\partial_\mu \Gamma^{\sigma}_{\rho\nu} + \frac{2}{3} \Gamma^{\sigma}_{\mu\beta} \Gamma^{\beta}_{\nu\rho}) \right].$$

- There exist a massive spin-2 modes.
- Chern-Simons term has third time derivative dependence $\Rightarrow$ ghost-free and unitary.

Chiral Gravity in Three-Dimension

- Consider conventional Einstein-Hilbert action (right-sign) in TMG with cosmological constant in AdS_3 spacetime

$$I = \int \sqrt{-g}\, d^3x \left[\frac{1}{2\kappa^2}(R - 2\Lambda) + \frac{1}{4\kappa^2\mu}\epsilon^{\lambda\mu\nu}\Gamma^{\rho}_{\lambda\sigma}(\partial_\mu\Gamma^{\sigma}_{\rho\nu} + \frac{2}{3}\Gamma^{\sigma}_{\mu\beta}\Gamma^{\beta}_{\nu\rho}) \right],$$

- The sign choice of the Einstein-Hilbert action is in order to make sure that the energy of the BTZ black hole is positive.
- 2+1 gravity on asymptotically AdS_3 spacetime with appropriate boundary condition can be descrided by a 2-dim conformal field teory (AdS_3/CFT_2).
- The chiral central charge is related to the coefficient of the Chern-Simons term and the total central charge $c_L + c_R = \frac{3l}{G}$.

- The diffeomorphism anomaly of the chiral central charge is $c_L - c_R = -\frac{3}{\mu G}$

$$\text{Solutions: } c_L = \frac{3l}{2G}(1 - \frac{1}{\mu l}), \quad c_R = \frac{3l}{2G}(1 + \frac{1}{\mu l}).$$

- Due to both central charges are non-negative $\Rightarrow \mu l \geq 1$.

Critical point in 3-dim gravity

$$\mu l = 1 \quad \Rightarrow \quad c_L = 0 \text{ and } c_R = \frac{3l}{G}.$$

- Only right-moving excitations $\Rightarrow$ chiral gravity.

New Massive Gravity (NMG)

- Consider the action

$$I = \frac{1}{2\kappa^2}\int \sqrt{-g}\, d^3x \left[-R + \frac{1}{m^2}K\right], \text{ where } K = R_{\mu\nu}R^{\mu\nu} - \frac{3}{8}R^2.$$

- The field equation $2m^2 G_{\mu\nu} + K_{\mu\nu} = 0$, where

$$\begin{aligned} K_{\mu\nu} &= 2\Box R_{\mu\nu} - \frac{1}{2}(\nabla_\mu\nabla_\nu R + g_{\mu\nu}\Box R) - 8R^\rho_\mu R_{\nu\rho} \\ &\quad + \frac{9}{2}RR_{\mu\nu} + g_{\mu\nu}[3R^{\rho\sigma}R_{\rho\sigma} - \frac{13}{8}R^2]. \end{aligned}$$

- Doing the linearisation, the field equation is equivalent to the Pauli-Fierz theory for massive spin-2 field.
- The Lagrangian is equivalent to the form

$$\mathcal{L} = \frac{1}{2\kappa^2}\sqrt{-g}\left[-R + f^{\mu\nu}G_{\mu\nu} - \frac{1}{4}m^2(f_{\mu\nu}f^{\mu\nu} - f^2)\right].$$

Review of Semiclassical Quantum Gravity

- The total action

$$I = I_g + I_m = \int \sqrt{-g}\, d^4x \frac{1}{2\kappa^2}(R - 2\Lambda) + I_m.$$

- Classical Einstein equation can be derived from above action

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

- The matter (include graviton) stress-tensor is regarded as a quantum expectation value in the strong gravity background

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda_B g_{\mu\nu} = 8\pi G_B \langle T_{\mu\nu}\rangle,$$

where

$$\langle T_{\mu\nu}\rangle = \frac{-2}{\sqrt{-g}}\frac{\delta W}{\delta g^{\mu\nu}}$$

with W knows as the *one-loop effective action* and $I = I_g + W$.

- The *one-loop effective action* is

$$W = \int \sqrt{-g}\, d^4x L_{eff}$$

with $L_{eff} = L_{ren} + L_{div}$.

- The divergence term
$L_{div} = const. + aR + bR^2 + cR^{\mu\nu}R_{\mu\nu} + dR^{\mu\nu\gamma\delta}R_{\mu\nu\gamma\delta}$
- Add high derivative term to the gravitational lagrangian

$$\begin{aligned}(I_g)_{ren} &= \int \sqrt{-g}\, d^4x \Big[\frac{1}{2\kappa^2}(R - 2\Lambda) \\ &\quad + \bar{b}R^{\mu\nu}R_{\mu\nu} + \bar{c}R^2 + \bar{d}R^{\mu\nu\gamma\delta}R_{\mu\nu\gamma\delta}\Big],\end{aligned}$$

- The action $I = I_g + W \to (I_g)_{ren} + W_{ren}$.

- The high derivative term give the contribution to the EoM are

$$
\begin{aligned}
{}^{(1)}H_{\mu\nu} &= \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\int\sqrt{-g}\,d^4x R^2, \\
{}^{(2)}H_{\mu\nu} &= \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\int\sqrt{-g}\,d^4x R^{\mu\nu}R_{\mu\nu}, \\
H_{\mu\nu} &= \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\int\sqrt{-g}\,d^4x R^{\mu\nu\gamma\delta}R_{\mu\nu\gamma\delta}.
\end{aligned}
$$

- The generalized Gauss-Bonnet theorem in 4-dim states that

$$
\int\sqrt{-g}\,d^4x(R^{\mu\nu\gamma\delta}R_{\mu\nu\gamma\delta}+R^2-4R^{\mu\nu}R_{\mu\nu})
$$

is a topological invariant which gives the relation

$$
\boxed{H_{\mu\nu}=-{}^{(1)}H_{\mu\nu}+4{}^{(2)}H_{\mu\nu}}.
$$

Renormalized gravitational action

$$I_g = \int \sqrt{-g}\, d^4x \left[\frac{1}{2\kappa^2}(R - 2\Lambda) + \alpha R^{\mu\nu} R_{\mu\nu} + \beta R^2 \right]$$

- Renormalized at one-loop quantum level but not unitary.
- The degree of freedom [K. S. Stelle (1977,1978)]

$$\left.\begin{array}{r} 2 \text{ massless spin-2} \\ 5 \text{ massive spin-2} \\ 1 \text{ massive scalar} \end{array}\right\} \Rightarrow 8 \text{ dgree of freedom.}$$

- Specific choice of the parameters

$$\begin{cases} \alpha = 0 \rightarrow \text{ massive spin-2 disappear,} \\ \alpha = -3\beta \rightarrow \text{ massive scalar disappear.} \end{cases}$$

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Critical Gravity

- Follow the chiral gravity in 3-dim to seek the limit (critical point) in which the massive spin-2 field becomes massless.
- Starting from the renormalized gravitational action

$$I_g = \frac{1}{2\kappa^2}\int \sqrt{-g}\, d^4x \left[(R-2\Lambda) + \alpha R^{\mu\nu}R_{\mu\nu} + \beta R^2\right].$$

- Using DeWitt's background field method $\tilde{g}_{\mu\nu} = g_{\mu\nu} + h_{\mu\nu}$ (1967) and take the background $g_{\mu\nu}$ to be AdS_4 ($\Lambda < 0$).

$$\Rightarrow \begin{cases} R_{\mu\nu} = \Lambda g_{\mu\nu}, \\ R = 4\Lambda, \\ R_{\mu\nu\rho\sigma} = \dfrac{\Lambda}{3}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}). \end{cases}$$

- The EoM from the action

$$
\begin{aligned}
& \mathcal{G}_{\mu\nu} + E_{\mu\nu} = 0, \\
\mathcal{G}_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu}, \\
E_{\mu\nu} &= 2\alpha(R_{\mu\rho} R_{\nu}{}^{\rho} - \frac{1}{4} R^{\rho\sigma} R_{\rho\sigma}) + 2\beta(R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu}) \\
& \quad + \alpha(\Box R_{\mu\nu} + \frac{1}{2} \Box R g_{\mu\nu} - 2\nabla_{\rho} \nabla_{(\mu} R_{\nu)}{}^{\rho}) \\
& \quad + 2\beta(g_{\mu\nu} \Box R - \nabla_{\mu} \nabla_{\nu} R).
\end{aligned}
$$

- Consider the linearisation around the background

$$
\begin{aligned}
\delta(\mathcal{G}_{\mu\nu} + E_{\mu\nu}) &= [1 + 2\Lambda(\alpha + 4\beta)] \mathcal{G}^{L}_{\mu\nu} \\
& \quad + \alpha \left[(\Box - \frac{2\Lambda}{3}) \mathcal{G}^{L}_{\mu\nu} - \frac{2\Lambda}{3} R^{L} g_{\mu\nu} \right] \\
& \quad + (\alpha + 2\beta)[-\nabla_{\mu} \nabla_{\nu} + g_{\mu\nu} \Box + \Lambda g_{\mu\nu}] R^{L}.
\end{aligned}
$$

- Impose the gauge condition inside the linearisation equation

$$\nabla^\mu h_{\mu\nu} = \nabla_\nu h.$$

- Take the trace to the linearisation equation and we have the scalar mode EoM

$$0 = g^{\mu\nu}\delta(\mathcal{G}_{\mu\nu} + E_{\mu\nu}) = \Lambda[h - 2(\alpha + 3\beta)\Box h].$$

- For specific condition $\alpha = -3\beta$ imply $h = 0$, the scalar disappeared.

- For tensor modes, we have the EoM

$$0 = \delta(\mathcal{G}_{\mu\nu} + E_{\mu\nu}) = \frac{3\beta}{2}\left(\Box - \frac{2\Lambda}{3}\right)\left(\Box - \frac{4\Lambda}{3} - \frac{1}{3\beta}\right) h_{\mu\nu},$$

where $h_{\mu\nu}$ is in the transverse traceless gauge

$$\nabla^{\mu} h_{\mu\nu} = 0, \quad g^{\mu\nu} h_{\mu\nu} = 0.$$

- The fourth-order equation

$$\begin{cases} \left(\Box - \frac{2\Lambda}{3}\right) h^{(m)}_{\mu\nu} = 0 & \text{massless graviton,} \\ \left(\Box - \frac{4\Lambda}{3} - \frac{1}{3\beta}\right) h^{(M)}_{\mu\nu} = 0 & \text{massive spin-2 modes.} \end{cases}$$

- Due to the stability for equation of spin-2 modes, the neaassary condition is

$$M^2 \equiv \frac{2\Lambda}{3} + \frac{1}{3\beta} \geq 0 \quad \Rightarrow \quad \beta \leq \frac{-1}{2\Lambda}.$$

Critical point in 4-dim gravity

$$\beta = \frac{-1}{2\Lambda}$$

- spin-2 massive modes become massless at critical point

$$\left(\Box - \frac{2\Lambda}{3}\right) h_{\mu\nu}^{(M)} = 0.$$

- There are two degenerate spin-2 massless modes gravitons.

Log Gravity

- There exist a new logarithmic modes in 4-dim critical gravity,

$$h^{log}_{\mu\nu} = f(t,\rho)h_{\mu\nu},$$

where $h_{\mu\nu}$ is the standard spin-2 massless modes and

$$f(t,\rho) = 2it + \log\sinh(2\rho) - \log\tanh(\rho).$$

- Consider D-dim gravity theory

$$I = \frac{1}{2\kappa^2}\int\sqrt{-g}\,d^Dx\left[\sigma R - 2\lambda m^2 + \frac{1}{m^2}G^{\mu\nu}S_{\mu\nu} + \frac{1}{m'^2}L_{GB}\right],$$

where L_{GB} is Gauss-Bonnet term.

- $S_{\mu\nu}$ is the D-dim Schouton tensor

$$S_{\mu\nu} = \frac{1}{D-2}\left(R_{\mu\nu} - \frac{1}{2(D-1)}Rg_{\mu\nu}\right).$$

- D=3 action is identically to the cosmological NMG.
- D=4 and $\sigma = 1$ action will reduced to the critical gravity.

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Extension

- D-dim critical gravity
- Adding scalar field?

  [Kyosuke Hirochi and Shin'ichi Nojiri, arXiv:1203.3301v1[hep-th]]

  The action of 4-dim critical gravity with a scalar field

$$I = \int \sqrt{-g}\, d^4x \left\{ \frac{1}{2\kappa^2}\left[R - 2\Lambda_0 + \alpha(R^{\mu\nu}R_{\mu\nu} - \frac{1}{3}R^2)\right] - \frac{1}{2}g^{\mu\nu}\nabla_\mu\nabla_\nu - \frac{1}{2}m^2\phi^2 + \gamma R\phi^2 - \frac{1}{4!}\xi\phi^4 \right\}$$

  which has additional parameter γ in the non-minimal coupple term to choose to have a critical condition and make $\Lambda > 0$ (dS_4 spacetime)

  $\Rightarrow$ Is it possible to describe the accelerating universe?

  $\Rightarrow$ Relate to the Gauss-Bonnet gravity?
- Relate to conformal gravity?
- Exist citical point in Einstein-Cartan theory?

End

- Thank you!